

CODE LAYOUT, READABILITY AND REUSABILITY

EDUGENIE – AI-POWERED LEARNING ASSISTANT

PROJECT ORGANIZATION

The project separates major responsibilities into individual Python files.

Benefits:

- Easier identification of each learning feature.
- Reduced responsibility per module.
- Easier maintenance.
- Easier debugging.
- Easier future extension.

REUSABILITY

The FastAPI backend can route different requests to dedicated modules.

The Gemini client provides a common integration point for cloud-based AI operations.

The frontend provides a shared interface for all learning functions.

READABILITY

Descriptive module names make the project structure understandable:

1. qna.py
2. explanation_module.py
3. quiz_module.py
4. summary_module.py
5. learning_path.py

CONFIGURATION SEPARATION

API credentials are separated from application source through environment variables.